

E(n) Equivariant Graph Neural Networks

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GitHub: <https://github.com/surya0925/E-n-Equivariant-Graph-Neural-Networks/tree/main>

Abstract

This paper introduces a new model to learn graph neural networks equivariant to rotations, translations, reflections and permutations called E(n)-Equivariant Graph Neural Networks (EGNNs). In contrast with existing methods, our work does not require computationally expensive higher-order representations in intermediate layers while it still achieves competitive or better performance. In addition, whereas existing methods are limited to equivariance on 3 dimensional spaces, our model is easily scaled to higher-dimensional spaces. We demonstrate the effectiveness of our method on dynamical systems modelling, representation learning in graph autoencoders and predicting molecular properties.

1 Introduction

E(n)-Equivariant Graph Neural Networks (EGNN) are designed to process graph or point cloud data while respecting geometric symmetries such as rotations, translations, reflections, and permutations. Many real-world datasets, including molecular structures and physical systems, naturally exhibit these symmetries. Incorporating them directly into the model architecture improves learning efficiency and helps models generalize better. While standard graph neural networks are inherently permutation equivariant, they do not preserve continuous Euclidean transformations like rotation and translation, which limits their effectiveness for geometric data.

EGNN addresses this limitation by using relative positional information between nodes. Instead of relying on complex representations such as spherical harmonics, the model uses simple operations based on pairwise distances and relative displacement vectors. These operations ensure that node features and coordinates update in a way that preserves the required symmetry properties. As a result, EGNN can capture geometric relationships efficiently while remaining computationally simple and applicable to spaces of any dimension.

Problem Formulation

Let $G = (V, E)$ where each node $v_i \in V$ has feature embedding $h_i \in \mathbb{R}^{n_f}$, coordinate $x_i \in \mathbb{R}^n$, and edges carry optional attributes $a_{ij} \in \mathbb{R}^d$. The Euclidean group $E(n)$ acts on coordinates as $x_i \mapsto Qx_i + g$ for $Q \in O(n)$, $g \in \mathbb{R}^n$. A function ϕ is $E(n)$ -**equivariant** if $\phi(Qx + g) = Q\phi(x) + g$, and $E(n)$ -**invariant** if $\phi(Qx + g) = \phi(x)$.

Problem. Given $G = (V, E)$ with $\{h_i\}$ and $\{x_i\}$, learn a message-passing layer $\{h_i^{(l+1)}, x_i^{(l+1)}\} = \text{EGCL}(\{h_i^{(l)}\}, \{x_i^{(l)}\}, E)$ such that

$$Qx^{(l+1)} + g, h^{(l+1)} = \text{EGCL}(Qx^{(l)} + g, h^{(l)}, E).$$

Equivariant Graph Convolutional Layer

The EGCL updates are ($C = \frac{1}{M-1}$, all ϕ are MLPs):

$$\begin{aligned} m_{ij} &= \phi_e\left(h_i^{(l)}, h_j^{(l)}, \|x_i^{(l)} - x_j^{(l)}\|^2, a_{ij}\right), & m_i &= \sum_{j \neq i} m_{ij}, \\ x_i^{(l+1)} &= x_i^{(l)} + C \sum_{j \neq i} (x_i^{(l)} - x_j^{(l)}) \phi_x(m_{ij}), & h_i^{(l+1)} &= \phi_h(h_i^{(l)}, m_i). \end{aligned}$$

Equivariance Properties

Observation (Distance Invariance). $\|(Qx_i + g) - (Qx_j + g)\|^2 = (x_i - x_j)^\top Q^\top Q (x_i - x_j) = \|x_i - x_j\|^2$, so $\|x_i - x_j\|^2$ is $E(n)$ -invariant. Consequently m_{ij} is invariant, the displacement sum in the coordinate update is equivariant, and $h^{(l+1)}$ is invariant — establishing the theorem above. A composition of L such layers is therefore also $E(n)$ -equivariant.

Observation (Permutation Equivariance). The symmetric aggregation $m_i = \sum_{j \neq i} m_{ij}$ ensures $\text{EGCL}(\{h_{\pi(i)}\}, \{x_{\pi(i)}\}, E) = \{h_{\pi(i)}^{(l+1)}, x_{\pi(i)}^{(l+1)}\}$ for any node permutation π .

2 Literature Review

Classical approaches to enforcing geometric symmetry in neural networks rely on hand-crafted features and data augmentation strategies. While effective in simple settings, these methods fail to generalize across arbitrary rotations, translations, and reflections. Equivariance-based architectures embed symmetry directly into the model, but earlier designs introduce significant computational overhead and are often limited to fixed-dimensional spaces.

2.1 Group Equivariant Neural Networks

Foundational work established group equivariance as a guiding principle for neural architecture design [3, 4, 22]. Convolutional Neural Networks exploit translation equivariance, while Graph Neural Networks (GNNs) [2, 5, 10] exploit permutation equivariance over graph-structured data. However, these methods are generally restricted to specific symmetry groups, do not jointly handle rotations, translations, and reflections, and lack explicit modelling of geometric node coordinates.

2.2 SE(3) and E(3) Equivariant Methods

Tensor Field Networks (TFN) [21] and SE(3)-Transformers [7] achieve SE(3) equivariance using spherical harmonics over higher-order representations, but this computation is expensive and cannot be scaled beyond three dimensions. Finzi et al. [6] use Lie Algebra kernel mappings, though outputs can be stochastic. Köhler et al. [14, 15] propose E(n)-equivariant flows for point clouds, but operate only on positional data without learnable feature dimensions.

2.3 Message Passing Networks for Molecular Data

Gilmer et al. [8] introduced the message passing framework for quantum chemistry, achieving permutation equivariance but not rotational or translational symmetry. SchNet [20] builds E(n)-invariant networks using relative distances, discarding directional information. DimeNet++ [12] recovers directionality through angles and spherical harmonics at added cost. Cormorant [1] and L1Net [18] incorporate SO(3) equivariance via higher-order representations, but remain computationally demanding.

2.4 Limitations in Prior Work

Across both classical and neural approaches, key challenges remain unresolved:

- Reliance on expensive higher-order representations such as spherical harmonics.
- Restriction of equivariance to three-dimensional spaces.
- Inability to jointly propagate node features and coordinates equivariantly.
- Absence of a unified, lightweight E(n)-equivariant framework.
- Limited data efficiency in non-equivariant models.

The EGNN framework addresses these limitations through an Equivariant Graph Convolutional Layer that updates node embeddings and coordinates using simple MLP operations on relative squared distances — achieving full E(n) equivariance without spherical harmonics, scaling to higher-dimensional spaces, and retaining the flexibility of standard GNNs.

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